

Multi-Reference EagleEye for Stellar Overdensity Detection in *Gaia*

Method definition, reference handling, and thresholding

Internal technical report — for project collaborators. Not a submission draft.

Draft of 20 September 2026. Section 9 (Calibration) is deliberately incomplete and is marked as such.

Contents

1	Purpose and scope	2
2	EagleEye in brief	3
2.1	The coin-flip statistic	3
2.2	IDE and repêchage	3
2.3	The injection background estimator	4
2.4	What is inherited and what is new	4
3	Data, coverage and configuration	4
3.1	The <i>Gaia</i> query	4
3.2	Coverage: the precondition for every number in this report	5
3.3	Feature construction	6
3.4	Configuration	6
4	The multi-reference construction	6
4.1	The 3×3 tiling	6
4.2	Robust scaling	7
4.3	Per-reference nulls	7
4.4	Γ as Fisher combination	8
5	Reference handling	8
5.1	Why a contaminated reference corrupts every test star	8
5.2	The saturation ceiling	9
5.3	The equalisation algorithm	9
5.4	K_M and the locality bound	10

6	Thresholding: the complete chain	11
6.1	Υ_j^* — the per-reference flagging threshold	11
6.2	Γ^* — the empirical threshold	11
6.3	A_Γ and clustering	12
6.4	$\hat{B}$ and $S/\sqrt{\hat{B}}$ per cluster	12
7	Recovery of known systems	12
7.1	Reticulum II	13
7.2	Boötes II	13
8	An unidentified overdensity in the Reticulum II field	15
8.1	Properties	15
8.2	Where the evidence is weak	16
8.3	The robustness scan	16
8.4	What has been ruled out, and what has not	17
9	Calibration — PARTIAL, FOR LATER COMPLETION	18
9.1	Nulls implemented	18
9.2	Stacking against pooling	18
9.3	What is missing	19
10	Open items	19
A	Hyperparameters	20
B	Parallelisation and determinism	20
C	Reserved	20
D	Figure manifest	20
	References	21

1 Purpose and scope

This report documents how we apply EagleEye (Springer et al. 2025) to *Gaia* DR3 phase-space data to search for localised stellar overdensities, and how every threshold in that pipeline is set. It is written so that a collaborator can reproduce a run from the configuration table in Section 3.4 and audit any flagged star back to the threshold that admitted it.

Three things are established here. First, the *multi-reference construction*: we compare a test field against eight empirical neighbouring sky tiles rather than against a background model, and combine the evidence into a single per-star statistic (Section 4). Second, *reference handling*: a concentrated overdensity sitting in one of those reference tiles corrupts the statistic for *every* test star, and we describe the mechanism, its detection, and its repair (Section 5). Third, *thresholding*: the complete chain from raw *Gaia* rows to a flagged cluster, with each threshold stated and

justified (Section 6). We then show the pipeline recovering two known dwarf galaxies (Section 7) and characterise one unidentified overdensity found alongside Reticulum II (Section 8).

One thing is **not** established. We do not yet have a calibrated false-alarm rate, and therefore we do not have a defensible detection threshold. Section 9 lists the machinery that exists for this and the measurements that have not been made. Until those exist, the per-cluster significance we report should be used for *ranking* candidates, not for deciding whether a candidate is real. This is stated in Section 6.4 where the statistic is defined, and again in Section 9.3.

Results in this report come from three runs, deliberately chosen to exercise different parts of the pipeline. They were made at **three different configurations**, and are not directly comparable with one another; Section 3.4 gives the parameters and says where the differences bite.

2 EagleEye in brief

This section is a summary of the parent method, included so that Section 4 onwards can be read without reference to the EagleEye paper. Readers familiar with the method should skip to Section 2.4.

2.1 The coin-flip statistic

Given a reference sample X and a test sample Y , EagleEye scores each point by the composition of its neighbourhood in the pooled set $U = X \cup Y$. For a test point Y_i , the membership of its k -th nearest neighbour is encoded as $b_i^k = 1$ if that neighbour lies in Y and 0 if it lies in X . Under the null hypothesis that the two samples are drawn from the same underlying density, the cumulative count $B(i, K) = \sum_{k \leq K} b_i^k$ behaves as a sequence of coin flips,

$$B(i, K) \sim \text{Binomial}(K, \hat{p}), \quad \hat{p} = \frac{n_Y}{n_X + n_Y}. \quad (1)$$

The exact null is hypergeometric, because neighbours are drawn without replacement; the binomial is an accurate approximation in the regime $K \ll n_X + n_Y$ that the method operates in.

The per-point anomaly score is the right-tail p -value of the observed count, maximised over neighbourhood rank:

$$\Upsilon_i = \max_{1 \leq K \leq K_M} [-\log \text{pval}(B_{\text{obs}}(i, K))]. \quad (2)$$

Maximising over K is what gives the statistic sensitivity across a range of anomaly scales without committing to one. Points with $\Upsilon_i \geq \Upsilon_+$ are flagged, where the threshold is calibrated by Monte Carlo (Section 6.1).

The parameter K_M is bounded above by the requirement that the neighbourhood be *local* — that the $X:Y$ ratio be approximately constant across it. The paper sets

$$K_M \leq 0.05 \min(n_X, n_Y). \quad (3)$$

The min matters, and is re-derived operationally in Section 5.4; an expression of the form $0.05(n_X + n_Y)$ appears elsewhere in the implementation but governs a different quantity, the neighbour re-fetch buffer.

2.2 IDE and repêchage

The flagged set Y_+ contains a halo of points whose neighbourhoods merely *overlap* a genuine anomaly. Two refinement stages separate signal from halo.

Iterative density equalisation (IDE) repeatedly removes the highest-scoring remaining point together with its nearest neighbours, and rescores, until nothing exceeds the threshold. The removed set $\hat{Y}_+$ is a representation of the local density *excess* — the mass that has to be taken out before the two samples become locally indistinguishable. It is smaller than Y_+ and is the quantity used wherever we need “the excess itself” rather than “everything the excess influenced”.

Repêchage clusters the flagged set using Density Peaks Advanced (d’Errico et al. 2021), as implemented in `dadapy` (Glielmo et al. 2022), then, within each cluster α , sets a cluster-specific threshold at a low quantile ($q = 10^{-2}$) of the Υ values of that cluster’s IDE members. Points exceeding it form the final anomaly set Y_α^{anom} .

The distinction matters later. Repêchage **deliberately re-admits the surrounding region** and so contains background by construction. That is what makes the repêchage set usable as a local background estimate (Section 6.4) and it is why reference cleaning removes the *pruned* set and not the repêched one (Section 5.3).

2.3 The injection background estimator

To estimate how much of an anomaly set is irreducible background, EagleEye moves each reference point into the test set one at a time, rescores it, and counts how many are themselves flagged. That injected count Y_α^{inj} , rescaled by the relative sizes of the two background components, estimates the background in the anomaly region:

$$\hat{B}_\alpha = \left| Y_\alpha^{\text{inj}} \right| \frac{n_Y - |\hat{Y}_+|}{n_X - |\hat{X}_+|}, \quad \Lambda_\alpha = \frac{|Y_\alpha^{\text{anom}}| - \hat{B}_\alpha}{\sqrt{\hat{B}_\alpha}}. \quad (4)$$

Λ_α is interpretable as an approximate z -score only in the large-background regime; outside it, it is a standardised measure of discrepancy strength. The estimator has a validity boundary: when the pruned mass exceeds the opposing sample, the numerator of $\hat{B}$ goes negative and the statistic is undefined. This is rare in the balanced configurations used here ($\hat{p} \approx 0.5$ throughout, Section 7) but common when references are pooled (Section 9.2).

Placeholder. A question concerning the implementation of this estimator is being raised directly with the EagleEye authors. It is deliberately not documented here, and Appendix C is reserved for it. Nothing in this report depends on its resolution: all runs presented are balanced ($n_X \approx n_Y$), which is the regime least sensitive to it.

2.4 What is inherited and what is new

This report circulates among EagleEye’s authors, so the boundary should be explicit.

One further item concerning the estimator implementation is being raised with the EagleEye authors directly and is deliberately not documented here.

3 Data, coverage and configuration

3.1 The *Gaia* query

All data are from `gaiadr3.gaia_source`. We retrieve `source_id`, `ra`, `dec`, `pmra`, `pmdec`, `phot_g_mean_mag`, `parallax`, `parallax_error` and `ruwe`, adding `bp_rp` when a colour–magnitude diagram is required. The query is a square box in RA/Dec about the field centre, with RA wrapping handled explicitly:

Inherited from EagleEye (Springer et al. 2025)	Introduced in this work
---	--------------------------------

Υ , the binomial k NN statistic IDE pruning; multimodal repêchage The injection background estimator Monte Carlo calibration of Υ^*	The eight-reference sky construction (§4.1) $\Gamma = \sum_j \Upsilon^{(j)}$ and its reading as Fisher combination (§4.4) Reference equalisation (§5.3) The empirical Γ^* bootstrap (§6.2) K_M resolution under the locality bound, including "auto" (§5.4) Parallelisation with verified bit-level determinism (Appendix B)
--	--

```

SELECT source_id, ra, dec, pmra, pmdec, phot_g_mean_mag,
       parallax, parallax_error, ruwe
FROM gaiadr3.gaia_source
WHERE (ra BETWEEN {ra_min} AND {ra_max})           - or (ra >= ra_min OR ra <= ra_max)
      AND (dec BETWEEN {dec_min} AND {dec_max})
      AND pmra BETWEEN -5 AND 5
      AND pmdec BETWEEN -5 AND 5
      AND (parallax - 3*parallax_error) < 0.0

```

Two cuts, each doing one job. The proper-motion box $|\mu| < 5 \text{ mas yr}^{-1}$ removes the nearby disc population, which would otherwise dominate the feature space, and bounds the PM features so that their scatter is set by the halo rather than by a handful of high-velocity foreground stars. The parallax cut $\varpi - 3\sigma_\varpi < 0$ retains only stars consistent with being distant, which is the relevant population for satellites at tens of kiloparsecs.

Two things are deliberately *not* cut, and both are open questions. We apply no RUWE cut, so poorly-behaved astrometric solutions remain in the sample; and we apply no extinction correction, so the stellar density field retains dust structure. The second is the more consequential, since a dust-driven density gradient is exactly the kind of large-scale structure the reference construction is meant to absorb (Section 4.1) and its residual is what Sections 5 and 8.3 test for.

3.2 Coverage: the precondition for every number in this report

The tiling routine does not complain when the 3×3 grid runs off the edge of the supplied table. It simply returns thinner outer tiles. This is the most dangerous failure mode in the pipeline, because a truncated reference tile contains too few stars, and test stars near the corresponding edge are then compared against a region with almost no reference — which the statistic reads as an overwhelming overdensity pinned to the tile boundary.

This is not hypothetical. A run on a $\pm 5^\circ$ table with a $(-1, -1)$ grid offset clipped **26 per cent of the western references’ RA extent** and manufactured a 390-star “cluster” at $S/\sqrt{\hat{B}} = 239$, in a patch that raw star counts show to be slightly *underdense*. The result was invalidated entirely.

Two guards now exist. A hard coverage check inside the run driver raises if the supplied table does not cover $\pm 3h$ about the window centre, where h is the tile half-width. And the query can be sized in advance: for a scan with box-size factors up to s_{max} and shift fraction ρ , the half-width guaranteed to cover every configuration is

$$\text{half} = h_0 (1 + s_{\text{max}}(3 + \rho)) + \text{margin}. \quad (5)$$

The operational rule, which we state as a rule because the incident above came from breaking it: **never run the pipeline on a table you did not size in advance.**

3.3 Feature construction

Each tile is represented in coordinates local to its own centre,

$$[\Delta\alpha \cos \delta_0, \Delta\delta, \mu_{\alpha*}, \mu_\delta], \quad (6)$$

with $\Delta\alpha$ wrapped to $(-180^\circ, 180^\circ]$. The cosine factor uses the *grid* centre declination δ_0 for all nine tiles rather than each tile's own declination. This is deliberate: a per-tile cosine would give each tile a slightly different angular scale, and the resulting feature spaces would not be commensurable when their Υ values are summed.

3.4 Configuration

The three runs in this report use three different configurations. They are collected here as the single source of truth; later sections refer to this table rather than restating parameters.

Table 1: Configuration of the three runs presented. The runs are *not* directly comparable.

Parameter	Boötes II (§7.2)	Reticulum II (§7.1, §8)	Ret II scan (§8.3)
Tile half-width h [deg]	1.0000	1.3333	0.667–1.5
K_M	50	100	"auto" (32–169)
p_{ext}	10^{-3}	5×10^{-4}	5×10^{-3}
n_{boot}	10	10	10
N_{null}	200 000	200 000	200 000
DPA Z	2.65	2.65	2.65
DBSCAN ϵ / min. samples	1.0 / 5	1.0 / 5	1.0 / 5
Reference equalisation	on , gate $S/\sqrt{\hat{B}} > 5$	off	on
Seed	42	42	42
n_Y	2 099	3 037	739–3 548
$\sum_j n_X^{(j)}$	16 517	23 897	—
Γ^*	43.79	52.06	35.2–45.1
Wall time	32 s	—	—

Two warnings follow from this table.

The runs are not directly comparable. Γ^* differs between Boötes II and Reticulum II (43.79 against 52.06) principally because p_{ext} and K_M differ, not because the fields differ. Any comparison of significance between Sections 7.1 and 7.2 is therefore a comparison of two different tests, and we do not make one.

The Reticulum II run has equalisation switched off. This is a limitation of that run, not a choice of method. We checked whether it matters: the nearest catalogued objects capable of contaminating a reference tile — NGC 1261, Horologium I and Reticulum III — all fall **outside** the 3×3 grid at $h = 1.3333^\circ$ (NGC 1261 lies 5.85° west of the grid centre in raw RA, against a grid half-extent of $4h = 4.0^\circ$ in the tiling coordinate). There is therefore no known contaminant in that field and no correctness problem. But it does mean the Reticulum II figures cannot illustrate equalisation, and the Boötes II run carries that load instead (Sections 5 and 7.2).

4 The multi-reference construction

4.1 The 3×3 tiling

We partition a square field into a 3×3 grid of tiles of half-width h . The central tile is the test sample $\mathcal{Y}$; the eight surrounding tiles are reference samples $\mathcal{X}^{(j)}$, $j = 1, \dots, 8$. Every reference is

real sky, observed by the same instrument, subject to the same selection function, at almost the same position on the sky.

The arrangement is chosen so that the references bracket the test tile symmetrically. Four antipodal pairs (E/W, N/S, and the two diagonals) surround $\mathcal{Y}$, so that a *linear* density gradient across the field contributes equally and oppositely to the members of each pair and cancels in the mean.

That cancellation is only partial, and it is important to say why immediately. $\Gamma = \sum_j \Upsilon^{(j)}$ is a sum of a non-negative, convex statistic. A linear gradient cancels in the *mean* of the underlying densities but not in the sum of the rectified scores: the tile on the sparser side contributes a positive Υ which the denser side cannot offset by contributing a negative one, because $\Upsilon \geq 0$. A strong gradient therefore leaves a one-sided residual along whichever edge of $\mathcal{Y}$ faces the sparser references. This is the motivation for the geometric tests in Section 8.3, and it is one of the artefacts the method-level nulls in Section 9.1 are designed to expose.

4.2 Robust scaling

Features are standardised by the median and median absolute deviation (MAD), fitted jointly across all nine tiles. Fitting on the pooled set rather than on $\mathcal{Y}$ alone matters because the scaling defines the metric in which nearest neighbours are found, and it must mean the same thing in the test tile and in every reference.

The reason the scaling cannot simply be refitted per configuration is that it is not scale-free. The positional MADs of a uniformly-filled tile scale linearly with the tile half-width, while the proper-motion MADs are pinned near 1.4 mas yr^{-1} by the ± 5 cut. Both runs confirm this directly:

Table 2: Measured MADs. The declination MAD is $0.50h$ in both runs; the RA MAD is $0.50 h \cos \delta_0$.

Run	h	$\text{MAD}(\Delta\alpha \cos \delta)$	$/h$	$\text{MAD}(\Delta\delta)$	$/h$	$\text{MAD}(\mu_{\alpha*})$	$\text{MAD}(\mu_\delta)$
Boötes II	1.000	0.4894	0.489	0.4985	0.499	1.425	1.457
Reticulum II	1.333	0.3943	0.296	0.6675	0.501	1.452	1.394

The declination MAD is $0.50h$ in both cases, as expected for a uniform tile of half-width h . The RA MAD is $0.50 h \cos \delta_0$ — $0.489 \approx 0.5 \cos(12.9^\circ)$ for Boötes II and $0.296 \approx 0.5 \cos(54.1^\circ)$ for Reticulum II. The positional MADs therefore vary by a factor of 1.7 between these two fields at fixed h purely through declination, while the PM MADs are constant to 4 per cent. The position-to-PM weighting of the Euclidean metric is consequently *not* a property of the method but of where and at what scale it is pointed.

Two consequences. When comparing configurations at different h — as the robustness scan does — the metric must be frozen from one configuration and reused, or the scan measures a changing distance function rather than a changing box. And because the DBSCAN radius ϵ is expressed in these scaled units, freezing the metric is also what makes a fixed ϵ correspond to a fixed angular size.

4.3 Per-reference nulls

Each reference has its own cardinality and therefore its own mixture proportion $\hat{p}_j = n_Y / (n_Y + n_X^{(j)})$. Each needs its own null distribution and its own threshold Υ_j^* . In the runs presented here the references are well balanced — $\hat{p}_j$ spans 0.493–0.515 for Boötes II and 0.490–0.519 for Reticulum II — so the eight thresholds are close, but they are computed separately regardless.

By contrast K_M is shared across all eight comparisons, and must be. Γ sums the $\Upsilon^{(j)}$, so they must be the same statistic; a per-reference K_M would give each term a different null *and* a different attainable ceiling (Section 5.2), so the sum would be weighted by tile size rather than by evidence.

4.4 Γ as Fisher combination

The combined statistic is

$$\Gamma_i = \sum_{j=1}^R \Upsilon_i^{(j)}, \quad R = 8. \quad (7)$$

Because $\Upsilon = -\log p$, this is $\Gamma_i = -\log \prod_j p_i^{(j)}$, which is exactly Fisher’s combined-probability statistic (Fisher 1932). If the R tests were independent with uniform p -values, 2Γ would be distributed as χ_{2R}^2 , equivalently $\Gamma \sim \text{Gamma}(R, 1)$.

They are not independent. All eight comparisons share the same test sample $\mathcal{Y}$, so the Poisson noise of $\mathcal{Y}$ is common to all of them and the $\Upsilon^{(j)}$ are positively correlated by construction. The nominal Fisher gain of $\sqrt{R}$ therefore overstates what stacking actually buys. This is not a small concern: Guth & Namjoo (2026) show that combining four correlated anomaly tests of the CMB yields a joint p -value around 3×10^{-8} that does not survive a proper treatment of the dependence. The same arithmetic error is available to us here, and the empirical null of Section 6.2 is what we use instead of the analytic one precisely to avoid it.

The degree of departure can be read off the fitted shape of the empirical Γ null. Matching a $\text{Gamma}(k, \theta)$ to its first two moments gives:

Run	mean	variance	$k_{\text{fit}} = \text{mean}^2/\text{var}$	θ	R
Boötes II	13.38	52.76	3.39	3.94	8
Reticulum II	15.70	61.39	4.02	3.91	8

so the null behaves like roughly four independent exponential contributions rather than eight.

Two cautions on reading that number. First, k_{fit} is measured on the *synthetic* uniform bootstrap null, so it characterises the method — the shared- $\mathcal{Y}$ correlation and the max-over- K selection — and is close to the same number for any field. It does not tell us whether *our* references differ from one another. Second, k_{fit} is not the same quantity as the effective reference count $R_{\text{eff}} = R/[1 + (R - 1)c]$ obtained from the measured pairwise correlation c of the $\Upsilon^{(j)}$ on real background stars. The two are related by $\text{mean}^2/\text{var} = R_{\text{eff}}(m^2/v)$ for marginal Υ mean m and variance v , and coincide only if Υ were exactly $\text{Exp}(1)$, which it is not.

R_{eff} is not reported here. The per-reference $\Upsilon^{(j)}$ arrays are not retained in the saved output of these runs, so the correlation cannot be recovered from them without re-running. This is a gap in what we can currently claim — it is the number that answers “how much independent evidence do eight references actually supply?”, and it is the honest counterpart to the two-algorithm corroboration convention used elsewhere in the satellite-search literature (e.g. Tan et al. 2026; Overdeck et al. 2026), which asserts agreement without quantifying it. Retaining `Upsilon_by_ref` in the run output is a one-line change and is listed in Section 10.

5 Reference handling

This section covers the failure mode that is specific to using real sky as a reference, and its repair. It is illustrated throughout by the Boötes II field, where the contaminant is not a hypothetical but a catalogued dwarf galaxy.

5.1 Why a contaminated reference corrupts every test star

Υ tests a *local* neighbourhood composition against a *global* mixture proportion $\hat{p} = n_Y/(n_Y + n_X)$. That comparison is valid as long as the two quantities correspond.

A reference tile that is uniformly denser than the test tile is handled correctly: the local $X:Y$ ratio is elevated everywhere by the same factor that elevates $\hat{p}$, and the two track each other. But a **concentrated** clump — a globular cluster, or, as here, a dwarf galaxy — is different. It inflates n_X , and therefore deflates $\hat{p}$, while leaving the local density at a randomly chosen field position completely unchanged. The global expectation and the local reality decouple, and *every* point of $\mathcal{Y}$ acquires an apparent excess of test-sample neighbours relative to a $\hat{p}$ that has been dragged down by structure nowhere near it.

The effect is a pedestal added uniformly to Υ across the whole test tile. Measured on NGC 1261 in a Reticulum II reference tile at a wider grid setting, $\hat{p}$ fell from 0.524 to 0.389 and a pedestal of +4.66 was added to Υ for all 7911 test stars. A pedestal of that size is comparable to the detection threshold itself.

5.2 The saturation ceiling

There is a second, more insidious consequence. Υ is the negative log of a binomial right-tail probability on a neighbourhood of at most K_M points, so the largest value it can attain — realised when every usable neighbour belongs to the test set — is

$$\Upsilon_{\max} = -(K_M - 1) \ln(1 - \hat{p}). \quad (8)$$

This ceiling depends only on K_M and $\hat{p}$. It is **independent of how large the contaminant is**. As a reference grows relative to the test tile, $\hat{p} \rightarrow 0$ and the ceiling collapses.

When it collapses below the detection threshold Υ^* , nothing in that reference can ever be flagged, and the cleaning pass returns “no clusters” — a result **indistinguishable from a clean reference**. This has been observed: for M3 sitting in a Boötes III reference tile at $K_M = 50$, $\Upsilon_{\max}$ and Υ^* were both 5.19, the reference-side flagged set was empty, and the most massive contaminant in the field was silently ignored.

The remedy is that the cleaning pass must run at its own K_M , chosen as large as the locality bound permits, rather than at the K_M chosen for the measurement. It must also refuse to proceed rather than return a null result when even that leaves no headroom, and we require $\Upsilon_{\max} - \Upsilon^* > 0.5$ before a pass is allowed to run.

5.3 The equalisation algorithm

Equalisation cuts localised overdensities out of the reference tiles, expressed entirely in the statistic’s own terms. For each reference independently, and iterating:

1. Recompute K_M from the current cardinalities as $\lfloor 0.05 \min(n_X^{(j)}, n_Y) \rfloor$, subject to a floor of 25 (below which the usable rank range $20 \leq K < K_M$ is empty).
2. Check the headroom $\Upsilon_{\max} - \Upsilon^*$ and raise if it is exhausted (Section 5.2).
3. Run the full EagleEye pass with roles reversed, so that overdensities *in the reference* are what is detected.
4. Gate the resulting reference-side clusters on $S/\sqrt{\hat{B}} > 5$.
5. Remove, for each gated cluster, its **IDE-pruned** set.

Step 5 is the substantive choice. We remove the pruned set and not the repêchaged set, because repêchage deliberately re-admits the surrounding region (Section 2.2) and so contains genuine background by construction. Removing only the excess mass excises the contaminant while leaving the local background in place, so the repaired reference still samples the true background at that position. Removing the repêchaged set instead would punch a hole in the reference.

There is deliberately **no cardinality target**. References of differing size are exactly what the multi-reference construction is built to handle, and a tile that is legitimately denser should remain

denser. The loop stops when EagleEye no longer finds anything in the reference, not when the tiles match.

A worked case. In the Boötes II field, the reference tiles are not blank sky: Boötes I, a kinematically confirmed dwarf with 204 members in the Battaglia et al. (2022) catalogue, lies 1.66° north and 0.51° east of the grid centre. At $h = 1.0^\circ$ this places it in cell 7, the north-central **reference** tile — outside $\mathcal{V}$, but squarely inside the background against which $\mathcal{V}$ is measured. Only 6 of its 204 members reach the test tile.

Equalisation found it. The results for all eight references:

Table 3: Reference equalisation in the Boötes II field. Seven references untouched; the one containing Boötes I repaired.

Reference cell	n before	n after	cut	passes	$S/\sqrt{\hat{B}}$	purity
0	2 079	2 079	0	0	—	—
1	2 159	2 159	0	0	—	—
2	2 152	2 152	0	0	—	—
3	2 123	2 123	0	0	—	—
5	2 043	2 043	0	0	—	—
6	1 974	1 974	0	0	—	—
7	2 102	2 005	97	1	26.02	0.839
8	1 982	1 982	0	0	—	—

Seven references were untouched; the one containing Boötes I was flagged at reference-side $S/\sqrt{\hat{B}} = 26.0$ and 97 stars were excised in a single pass, at an estimated purity of 0.84. The repaired tile is drawn in Figure 2, where the excised stars are shown separately.

This is the clearest available demonstration that the failure mode is real, that it arises in the ordinary course of pointing at a field with known neighbours rather than in a contrived case, and that the method detects its own contamination without being told where to look.

5.4 K_M and the locality bound

Υ assumes the $X:Y$ ratio is constant inside the K -neighbourhood. It is not — two adjacent sky tiles have a slowly varying ratio — and the neighbourhood radius grows like $k^{1/d}$, so the larger K_M is, the more of that variation the statistic swallows and reports as signal. On a synthetic field with a mild ratio gradient and no injected anomaly, the false-positive rate as a multiple of the nominal p_{ext} is:

Table 4: False-positive inflation as a multiple of nominal p_{ext} , measured on a synthetic gradient field with no anomaly.

K_M	25	50	100	300	600
$d = 2$	$2.0\times$	$6.7\times$	$12.5\times$	$46.0\times$	$118\times$
$d = 4$	$1.3\times$	$1.8\times$	$2.0\times$	$2.2\times$	$6.8\times$

The features here are four-dimensional, where the inflation is far milder — a factor of two out to $K_M = 300$. This is why setting K_M to the locality ceiling is a reasonable default rather than a reckless one.

Pushing against that, larger K_M is broadly *better* for sensitivity: the threshold Υ^* grows only logarithmically with K_M while the ceiling Υ_{max} grows linearly, and a diffuse anomaly accumulates its excess over many neighbours and may be invisible at small K_M regardless of how many stars

it contains. The "auto" setting resolves K_M to the largest value the locality bound permits, and it is resolved *after* equalisation — resolving it before would let a contaminated tile's inflated cardinality set the bound.

The consequences of the bound are visible in the robustness scan of Section 8.3, where shrinking the box by a factor 0.67 reduces n_Y to 739 and drives K_M^{auto} down to 32, close to the floor of 25.

6 Thresholding: the complete chain

This section states every threshold between raw *Gaia* rows and a flagged cluster. The chain is

$$\text{rows} \rightarrow \mathcal{Y}, \mathcal{X}^{(j)} \rightarrow \Upsilon_j^* \rightarrow \Upsilon_i^{(j)} \rightarrow \Gamma_i \rightarrow \Gamma^* \rightarrow A_\Gamma \rightarrow \text{DBSCAN} \rightarrow \hat{B}_w \rightarrow S/\sqrt{\hat{B}}.$$

6.1 Υ_j^* — the per-reference flagging threshold

Because Υ_i is a maximum over K , its null distribution is not the tail of any single binomial and must be obtained by simulation. For each reference we draw $N = 200\,000$ independent Bernoulli sequences of length K_M at probability $\hat{p}_j$, compute Υ for each exactly as for a real point, and take

$$\Upsilon_j^* = \text{quantile}(\{\Upsilon\}, 1 - p_{\text{ext}}). \quad (9)$$

One threshold per reference, because $\hat{p}_j$ differs.

p_{ext} **is a per-point exceedance level.** It is not a cluster-level or field-level error rate, and it does not describe how often the pipeline produces a spurious *cluster*. This is the most commonly misread parameter in the pipeline and it is the reason Section 9 exists: converting a per-point level into a statement about candidate reliability requires the false-alarm measurements we have not yet made.

6.2 Γ^* — the empirical threshold

The threshold on the combined statistic is obtained by bootstrap. For each of $n_{\text{boot}} = 10$ realisations we draw Y_b and $X_b^{(j)}$ uniformly on $[0, 1]^4$ **at the realised cardinalities of the actual run**, push them through the identical multi-reference pipeline, and collect the resulting $\Gamma_i^{(b)}$. Pooling across realisations and points gives the null sample, and

$$\Gamma^* = \text{quantile}(\Gamma_{\text{null}}, 1 - p_{\text{ext}}). \quad (10)$$

For the two runs here this gives null samples of 20 990 and 30 370 values respectively, and:

Table 5: Empirical Γ null quantiles.

Run	$q_{0.9}$	$q_{0.99}$	$q_{0.999}$	$q_{0.9999}$	Γ^* used	$\max_i \Gamma_i$
Boötes II	23.32	35.06	43.79	49.28	43.79 ($p_{\text{ext}} = 10^{-3}$)	76.22
Reticulum II	26.24	38.80	49.56	57.46	52.06 ($p_{\text{ext}} = 5 \times 10^{-4}$)	253.08

The empirical route is used rather than the analytic $\text{Gamma}(k_{\text{fit}}, \theta)$ fit because the true law is a sum of correlated maxima-of-exponentials whose tail the max-over- K selection reshapes. The fitted form tracks the null well to about the 10^{-2} level and then runs high — by roughly 9 per cent at $p_{\text{ext}} = 10^{-3}$ and 15 per cent at 10^{-4} . Substituting it would buy better run-to-run stability at the cost of a larger bias, in the conservative direction, discarding real anomalies.

The tail is where n_{boot} matters. At $n_{\text{boot}} = 10$ and $n_Y \approx 2\,000$ – $3\,000$ the null sample contains 2 – 3×10^4 values, so a quantile at $p_{\text{ext}} = 10^{-3}$ is estimated from roughly 20–30 points and at

5×10^{-4} from roughly 10–15. These are thin, and Γ^* carries a corresponding sampling uncertainty that we have not propagated. Running at smaller p_{ext} without increasing n_{boot} is not meaningful.

6.3 A_Γ and clustering

Points exceeding the threshold form the anomalous set $A_\Gamma = \{i : \Gamma_i > \Gamma^*\}$, which is then partitioned by DBSCAN (Ester et al. 1996) in the full four-dimensional scaled feature space with $\epsilon = 1.0$ and min. samples = 5. Clustering in the same space in which the statistic was computed — rather than on the sky alone — means a group must be coherent in position *and* proper motion to survive.

Two properties of this step should be kept in mind. Because ϵ is in MAD units, its physical meaning is fixed only if the metric is frozen (Section 4.2). And DBSCAN’s noise label is retained rather than suppressed: points above threshold that join no cluster are reported as unclustered and excluded from the cluster statistics. For Boötes II, 20 points exceeded Γ^* and 18 formed a single cluster, leaving 2 unclustered; for Reticulum II, 111 points exceeded threshold and all 111 were assigned to two clusters with no unclustered remainder.

6.4 $\hat{B}$ and $S/\sqrt{\hat{B}}$ per cluster

Each Γ cluster needs a background estimate, and the estimate available is per-reference: the repêchage stage of each of the eight comparisons produces its own clusters, each with its own $\hat{B}$ from the injection estimator of Section 2.3. We combine them by overlap weighting. For a Γ cluster α , let $o_\alpha^{(j)}$ be the number of stars it shares with the overlapping repêchage cluster in reference j . Then

$$\hat{B}_\alpha = \frac{\sum_j o_\alpha^{(j)} \hat{B}_\alpha^{(j)}}{\sum_j o_\alpha^{(j)}}, \quad S = |A_\Gamma^{(\alpha)}|, \quad \text{reported as } S/\sqrt{\hat{B}_\alpha}. \quad (11)$$

Only references contributing a finite, positive $\hat{B}$ and a finite, positive significance are included, so the number of contributing references is itself diagnostic: a cluster supported by all eight is a different object from one supported by two. In Section 7 the Reticulum II detection draws on all 8 references while the unidentified cluster draws on 2, and the Boötes II detection on 5.

What this number is, and is not. It is a local estimate of excess over background, and it is a useful *ranking* statistic. It is not a calibrated significance. It takes no account of how many independent places in the tile could have produced a comparable clump by chance, so it cannot be read as a trials-corrected detection significance, and it should not be quoted as a number of sigma. Converting it into such a statement is what Section 9 is for, and that work is not done.

7 Recovery of known systems

We show the pipeline recovering two kinematically confirmed dwarf galaxies from the Drlica-Wagner et al. (2020) catalogue, using membership from Battaglia et al. (2022) at $P_{\text{memb}} > 0.5$ as ground truth. Members are matched to our *Gaia* DR3 rows by `source_id`, so the comparison involves no positional tolerance.

For context, an earlier batch application of this pipeline to all 30 class-4 dwarfs with usable fields recovered 21 at a true-positive rate above 0.5 and 9 not at all, with no intermediate outcomes. That batch used a different configuration from anything in this report (wider tiles, $K_M = 300$, $p_{\text{ext}} = 10^{-5}$, no equalisation) and its per-system numbers are therefore not comparable with those below; Boötes II in particular is re-run here under the configuration of Section 3.4 and the result differs. We note the aggregate only as an indication of scale and do not draw on the batch further.

7.1 Reticulum II

Reticulum II has 75 Battaglia members, all of which fall inside the test tile. The run returns **two** Γ clusters from 111 points above threshold, with no unclustered remainder.

Table 6: The two Γ clusters in the Reticulum II field.

	cluster 1	cluster 0
n	89	22
$\hat{B}$ (overlap-weighted)	19.96	5.34
$S/\sqrt{\hat{B}}$	19.92	9.52
Contributing references	8 of 8	2 of 8
Best catalogue match	Reticulum II	<i>unidentified</i>
Members recovered	62 of 75	0
TPR	0.827	—
Purity against known members	0.697	0.000
Centroid (α, δ)	(53.908, -54.036)	(52.884, -55.027)
Separation from catalogue position	0.017°	1.142°
Mean proper motion [mas yr $^{-1}$]	(+2.42, -1.44)	(-0.40 , -0.01)
PM dispersion [mas yr $^{-1}$]	(0.38, 0.44)	(0.38, 0.35)
Median G	19.68	20.29

Cluster 1 is Reticulum II. It recovers 62 of 75 known members, its centroid lies $1.0'$ from the catalogue position, and its mean proper motion matches the published value. It is supported by all eight references, which is what a genuine, strong overdensity should look like: no choice of background makes it go away. Of its 89 members, 27 are not in the Battaglia list — a purity of 0.70 — which is the expected behaviour of the repêchage stage, which admits the surrounding region by construction (Section 2.2), and is not by itself evidence of contamination.

Cluster 0 matches nothing in the catalogue and is the subject of Section 8.

7.2 Boötes II

Boötes II has 22 Battaglia members, all inside the test tile. The run returns a single Γ cluster from 20 points above threshold, with 2 unclustered.

Table 7: The single Γ cluster in the Boötes II field.

	value
n	18
$\hat{B}$ (overlap-weighted)	3.93
$S/\sqrt{\hat{B}}$	9.08
Contributing references	5 of 8
Best catalogue match	Boötes II
Members recovered	14 of 22
TPR	0.636
Purity against known members	0.778
Centroid (α, δ)	(209.537, $+12.861$)
Separation from catalogue position	0.023°
Mean proper motion [mas yr $^{-1}$]	(-2.54 , -0.36)

Boötes II is a considerably harder target than Reticulum II — 22 catalogued members against 75 — and the recovery is correspondingly less complete: 14 of 22 members, at $S/\sqrt{\hat{B}} = 9.08$, supported by 5 of the 8 references rather than all 8. The centroid is $1.4'$ from the catalogue position. Of the 18 cluster members, 14 are known members, so the cluster is 78 per cent pure.

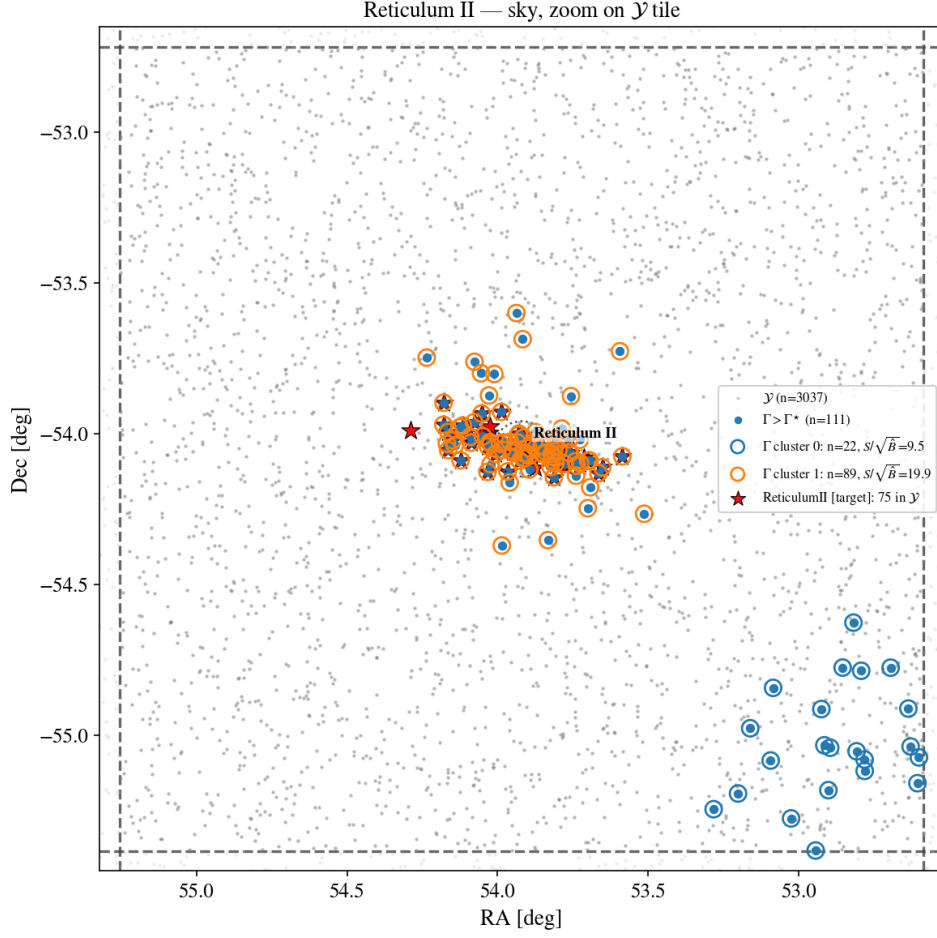


Figure 1: The Reticulum II test tile, with the 111 points above Γ^* marked and the two clusters distinguished. Reticulum II’s Battaglia members are overplotted as red stars. Cluster 1 sits on the target at the tile centre; cluster 0 occupies the south-western corner. Proper-motion counterpart: [figures/fig1b_reticulumII_pm.png](#).

The more instructive aspect of this field is what is happening in the *references*. Boötes I lies 1.66° north of the grid centre, placing it inside the north-central reference tile (Section 5.3), with 204 catalogued members of which only 6 reach the test tile. Equalisation flagged that tile at reference-side $S/\sqrt{\hat{B}} = 26.0$ and removed 97 stars; the other seven references were untouched. Without that repair, Boötes I’s 204 members would have inflated n_X for one of the eight comparisons, depressing $\hat{p}$ and adding a pedestal to $\Upsilon^{(j)}$ for all 2 099 test stars (Section 5.1).

This is the case the multi-reference construction has to handle in order to be usable: any field chosen because it contains one known satellite is reasonably likely to contain another within a few degrees.

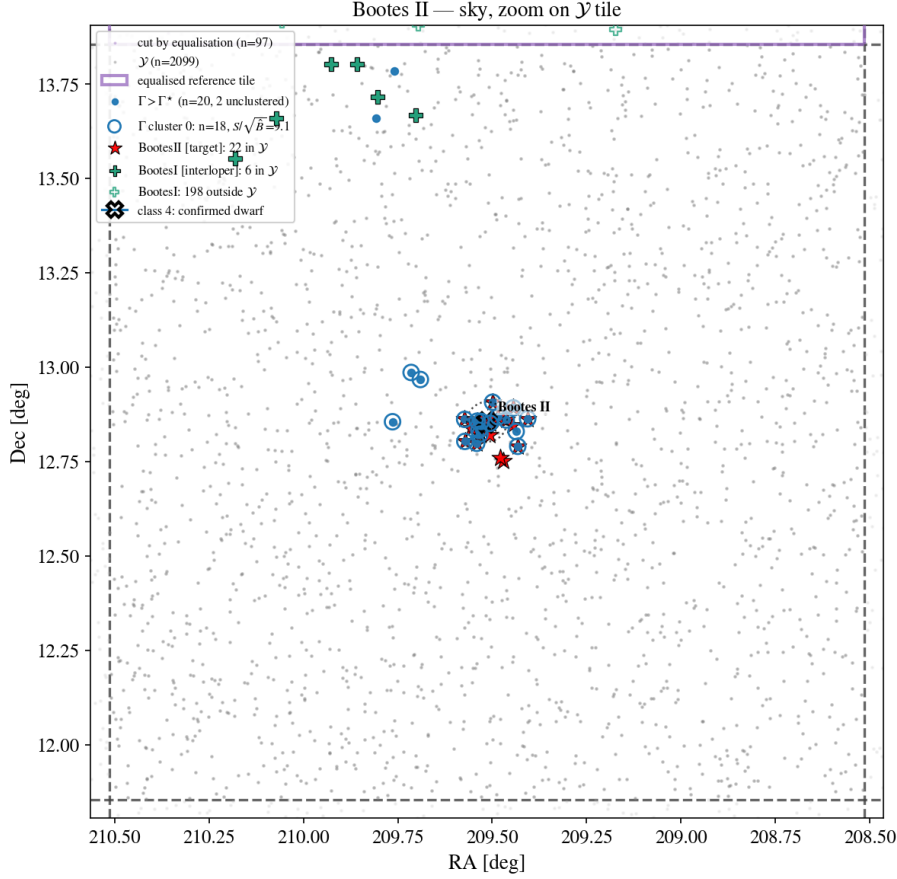


Figure 2: The Boötes II test tile with the recovered cluster, the Battaglia members of Boötes II (targets) and of Boötes I (interlopers, 6 inside the tile and 198 outside), and the repaired reference tile drawn at the top with its 97 excised stars marked. Colour–magnitude counterpart: `figures/fig2b_bootesII_cmd.png`.

8 An unidentified overdensity in the Reticulum II field

Cluster 0 of Section 7.1 matches no catalogued system. This section characterises it. We present it as a **candidate requiring follow-up, not as a detection**, and the evidence below is mixed.

8.1 Properties

The cluster contains 22 stars at $(\alpha, \delta) = (52.884, -55.027)$, 1.14° from Reticulum II. It is compact and round — its members span 0.26 and 0.28 of the tile in RA and declination respectively, an aspect ratio of 1.1:1, which is essentially identical to Reticulum II’s own footprint in the same run (0.27×0.29 , 1.1:1).

Its proper motion is tightly concentrated at $(-0.40, -0.01) \text{ mas yr}^{-1}$ with dispersions $(0.38, 0.35)$, again comparable to Reticulum II's $(0.38, 0.44)$. Its members are about 0.6 mag fainter in median G (20.29 against 19.68).

It is not associated with Reticulum II. The two proper motions differ by 2.9 mas yr^{-1} , an order of magnitude larger than either dispersion. At Reticulum II's distance of roughly 30 kpc that corresponds to a relative transverse velocity of several hundred km s^{-1} , far in excess of anything bound to it.

An overdensity at a consistent position has been seen in this field at other configurations, with somewhat different membership — 27 stars at $(52.881, -54.936)$ in an earlier run. The right ascensions agree to 0.003° and the declinations differ by 0.09° . We take these to be the same feature seen under different tilings, but we have not established that by matching `source_id` between runs, and it should be checked.

8.2 Where the evidence is weak

Two things should be stated plainly before the robustness results.

It is only supported by 2 of the 8 references, against 8 of 8 for Reticulum II in the same run. The overlap-weighted background is therefore built from two repêchage clusters rather than eight, and $\hat{B} = 5.34$ is correspondingly less well determined than Reticulum II's 19.96.

It touches the tile boundary. Its members come within $0.011h$ of the western edge and $0.001h$ of the southern edge — that is, they reach the corner of the test tile. This matters in two ways. Its true extent may be larger than measured, because any members beyond the boundary are not in $\mathcal{Y}$ at all; and worse, those members would be sitting in the adjacent reference tiles, where they would act to suppress the very feature we are trying to measure. The 22 stars should be read as a lower bound on the membership.

It is *not*, however, the classic tiling artefact. That signature is a thin strip lying flush along an edge and spanning the full width of the tile — high aspect ratio, large span, centroid pushed to the boundary. This cluster has an aspect ratio of 1.1 and spans about a quarter of the tile in each direction. It is a compact clump that happens to lie near a corner, which is a different thing, and the standard geometric test does not flag it.

8.3 The robustness scan

To separate a feature locked to the *sky* from one locked to the *tiling*, the pipeline is re-run on eight grids anchored on the anomaly itself: the nominal grid, four shifts pushing the anomaly toward each tile edge, two changes of box size, and one diagonal shift. A real object stays at a fixed sky position while the grid moves under it; an artefact follows the grid.

The reference set A_0 for this scan contains 28 stars.

Table 8: The eight-configuration robustness scan. Six of seven perturbed configurations recover the feature; the failure is the smallest box, where the locality bound drives K_M to 32.

Configuration	h	n_Y	K_M	Γ^*	clusters	recovered	overlap	recovery	Jaccard	$S/\sqrt{\hat{B}}$
nominal	1.000	1 627	75	39.75	3	yes	27	0.96	0.93	8.63
shiftE	1.000	1 659	75	40.43	2	yes	22	0.79	0.79	6.76
shiftW	1.000	1 585	74	40.53	1	yes	26	0.93	0.84	6.53
shiftN	1.000	1 577	74	40.65	2	yes	24	0.86	0.80	6.63
shiftS	1.000	1 627	75	39.94	2	yes	27	0.96	0.79	7.78
scale $\times 0.67$	0.667	739	32	35.22	0	no	0	0.00	0.00	—
scale $\times 1.50$	1.500	3 548	169	45.09	3	yes	23	0.82	0.64	6.98
diagonal	1.000	1 596	74	39.01	2	yes	27	0.96	0.84	9.00

The formal verdict is ROBUST. Six of the seven perturbed configurations recover the feature (0.86, against a requirement of 0.75); the median Jaccard index among recoveries is 0.80, against a requirement of 0.5; and the significance ranges over 6.53–9.00, a ratio of 1.38, against a permitted ratio of 3.0. The feature is locked to the sky, not to the tiling.

The single failure is the smallest box, and it is explained by the method rather than being anomalous. Shrinking the tile to $h = 0.667^\circ$ reduces n_Y to 739, which through the locality bound of Section 5.4 drives K_M^{auto} down to 32 — close to the floor of 25 below which no test exists at all. The attainable Υ_{max} falls with it, and a feature of this strength can no longer clear the threshold: 18 per cent of A_0 remained above Γ^* but no cluster formed. This is the documented behaviour of a shrinking box, not evidence against the feature. It does mean that a single significance threshold applied across all eight configurations would be biased against the small ones.

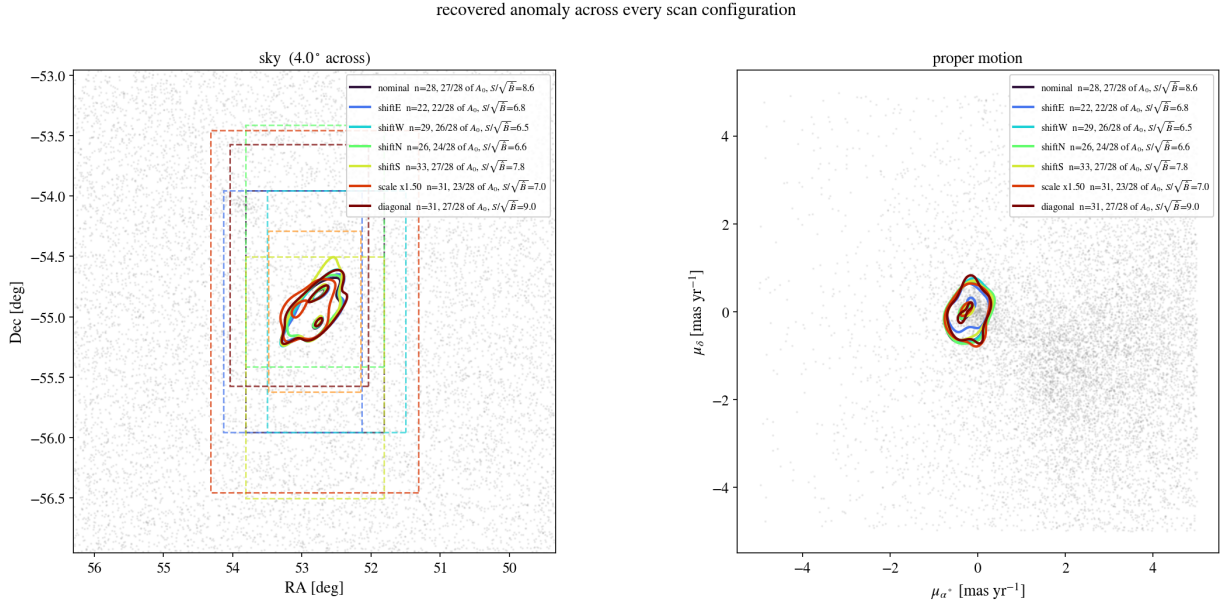


Figure 3: Left: the sky, with the recovered anomaly’s contour drawn for each configuration and each configuration’s test-tile boundary shown dashed. The contours coincide while the boundaries move, which is the signature of a sky-locked feature. Right: the same in proper-motion space, where Reticulum II is visible as the field overdensity near $(+2.4, -1.4)$ mas yr $^{-1}$ and the anomaly contours sit separately near $(-0.4, 0.0)$. The $\times 0.67$ configuration is absent from both panels because it returned no cluster.

8.4 What has been ruled out, and what has not

Two interpretations have been assessed and set aside. A stellar-wake origin — an overdensity of halo field stars trailing a dark perturber — is not supported: the shear from any plausible perturber at this distance is orders of magnitude below what is measurable with these data, and the proper motions are in any case incompatible with a common origin with Reticulum II. A boundedness argument is also unavailable: the test has no discriminating power at 30 kpc, as demonstrated by running it on Reticulum II itself, a known bound system, where it fails identically. Both are reported so they are not attempted again.

What has **not** been established is the alternative that matters: that 22 stars this tightly grouped in proper motion could not arise by chance within a field whose proper-motion distribution is itself strongly clustered. Answering that requires the calibration of Section 9, and until it exists the candidate stands on the robustness scan alone — which shows the feature is real in the data and not an artefact of the tiling, but not that it is a physical system.

9 Calibration — PARTIAL, FOR LATER COMPLETION

This section is deliberately incomplete. It records the machinery that exists and the measurements that have not been made. **No threshold recommendation should be taken from it.**

9.1 Nulls implemented

Four nulls exist, in increasing order of realism. Each answers a different question and none answers all of them.

Proper-motion scramble. Permute the $(\mu_{\alpha*}, \mu_{\delta})$ pairs among stars in the window. Positions are untouched, so the spatial density field and all its gradients survive; the joint PM distribution survives because the pairs are only relabelled; what is destroyed is the position–PM coherence that a genuine comoving group has. Applied to the Reticulum II-field candidate this gave an observed $S/\sqrt{\hat{B}} = 7.05$ against a maximum of 3.54 over 12 blank trials, $p < 0.08$. Twelve trials is too few to say more. This null can only answer “could the pipeline have manufactured this from the marginals alone?” — it cannot distinguish a genuine chance density from a physical system.

Off-position. Run the identical pipeline at positions elsewhere on the sky and read the empirical distribution of $S/\sqrt{\hat{B}}$ off blank sky. This manipulates nothing and is the classic off-source measurement. Two hardening steps were necessary after a first pass was found unsound: the exclusion radius around known objects must be the full $3h$ half-width of the grid rather than the test tile, because an object just outside $\mathcal{V}$ sits in a *reference* and does exactly the damage described in Section 5.1; and the DW20 catalogue contains no globular clusters, so those must be excluded using a separate catalogue (Vasiliev & Baumgardt 2021). Off-positions are also matched in stellar density to the target field, since surface density varies strongly with Galactic latitude.

Smooth mock. Resample the field with replacement and jitter by a kernel wider than any clump but narrower than the field’s real structure. Below the bandwidth the generating density is smooth by construction, so anything recovered is manufactured by the method, while the large-scale gradient and tile-to-tile density differences survive. This isolates method artefacts — tile-edge effects, the rectification described in Section 4.1, DBSCAN scale coupling — without appealing to what is or is not in the sky.

Scan-level persistence. The one that matters for flagging policy. A marginal cluster in a single configuration means little; the same cluster reappearing at the same sky position across all eight configurations of Section 8.3 is a different claim, and carries power the single-run significance does not express. Measuring how often blank fields produce such a persistent group gives the number to flag on. The machinery is written and parallelised. **No number has been produced.**

9.2 Stacking against pooling

There is an alternative use of eight references: concatenate them into a single pooled reference and run the standard single-comparison pipeline once. This is a genuinely different statistic, with $\hat{p} \approx 1/9$ rather than $\approx 1/2$, and it is worth recording as a limitation of stacking that the two are not uniformly ordered.

On synthetic fields built with the real cardinalities of a *Gaia* tile and deliberately lumpy, unequal-sized references, recovery rates were:

Compact anomalies favour pooling and diffuse anomalies favour stacking. The mechanism is understandable: stacking is a persistence measure and suppresses variance on a signal present in every comparison, which is what a diffuse excess looks like; but a $\hat{p}$ pedestal arising from reference lumpiness is additive and undiluted under stacking, whereas pooling averages it away.

Anomaly	Stacked	Pooled
Compact ($w = 0.035$, $N = 10$)	0.25	0.95
Diffuse ($w = 0.085$, $N = 20$)	0.43	0.23

These numbers rest on two trials per grid cell and should not be relied on. The direction of the effect is suggestive and the mechanism is plausible, but the statistics are thin. We record it here as a caveat attached to the stacked approach used throughout this report, and note that the candidate of Section 8 — compact, 22 stars — sits in the regime where stacking performed worse in this test. Running the pooled comparison on the real field is a cheap cross-check and has not been done.

9.3 What is missing

- The scan-level persistence false-alarm rate. Staged and parallelised; **no number**.
- The off-position null re-run under the hardened exclusion logic.
- The scramble null beyond 12 trials.
- Whether the stacking/pooling crossover survives more than two trials per cell.
- A pooled cross-check on the Reticulum II field.

Consequence. Until these exist there is no defensible detection threshold, and $S/\sqrt{\hat{B}}$ should be used to rank candidates, not to decide them. The candidate of Section 8 is reported on that basis.

10 Open items

Code.

- The per-reference $\Upsilon^{(j)}$ arrays are not retained in the saved run output, so R_{eff} cannot be computed after the fact (Section 4.4). Retaining `Upsilon_by_ref` is a one-line change and should be made before the next production runs.
- The injection estimator divides by zero rather than returning NaN when the injected count is zero.
- One further item concerning that estimator is pending discussion with the EagleEye authors (Sections 2.3 and Appendix C).
- The scan-overlay plotting has unguarded RA-wrap exposure.
- The null-calibration routine is unseeded.
- **Nothing is committed.** The EagleEye working copy carries uncommitted changes to `EagleEye.py`, `utils_EE.py`, `utils_systematics.py` and `test_multiBkg_ordered.py`. This should be resolved before anything else touches those files, since the runs in this report depend on that working state and it is not currently reproducible from version control.

Analysis.

- Confirm by `source_id` that the 22-star cluster of Section 8 and the 27-star cluster seen at other configurations are the same feature.
- Re-run Reticulum II with equalisation enabled, for consistency with Section 7.2 even though no contaminant is expected (Section 3.4).
- Run the pooled cross-check on the Reticulum II field (Section 9.2).
- Establish what distinguishes the fields in which the method recovers a known dwarf from those in which it recovers nothing. The batch outcome was strikingly bimodal and the cause is not known.

A Hyperparameters

Consolidated from Section 3.4, with justification.

h , tile half-width. Sets n_Y , and through the locality bound sets the achievable K_M and hence the achievable significance (Sections 5.4 and 8.3). Smaller tiles are not simply “more local”: they are also less sensitive.

K_M . Bounded above by $0.05 \min(n_X, n_Y)$ for locality and below by 25, since the usable rank range is $20 \leq K < K_M$. Larger is broadly better for sensitivity because Υ^* grows logarithmically while $\Upsilon_{\max}$ grows linearly; the cost is gradient leakage, which in four dimensions is mild (Section 5.4). Must be shared across all eight references (Section 4.3), and resolved after equalisation.

p_{ext} . A per-point exceedance level, not a cluster or field error rate (Section 6.1). Smaller tiles need a looser value: they carry a smaller multiple-comparison burden, so the same physical excess sits at a less extreme quantile, and simultaneously a lower $\Upsilon_{\max}$ ceiling. A diffuse overdensity is the case that most needs the looser threshold, since it produces many mildly elevated points rather than a few extreme ones.

n_{boot} . Controls how well the tail of the Γ null is resolved. At $n_{\text{boot}} = 10$ the quantile at $p_{\text{ext}} = 10^{-3}$ rests on 20–30 null values (Section 6.2).

DBSCAN ϵ , min. samples. In scaled units, so their physical meaning depends on the metric being frozen (Section 4.2).

DPA Z . Cluster-splitting parameter for the repêchage stage, left at the EagleEye default.

B Parallelisation and determinism

Three levels of parallelism exist and their product must stay within the core count: configurations $\times$ references $\times$ inner k NN threads.

Only one backend works. Forking breaks OpenBLAS, because numpy initialises its thread pool in the parent and the child inherits descriptors for threads that do not exist. Spawning re-imports `__main__`, which is unusable from a notebook. Loky with the inner thread limit pinned to one works, and is what is used. Note also that joblib disables nested parallelism, so requesting eight configurations *and* eight references does not give sixty-four workers — the inner level silently degrades.

Determinism required explicit care. Reseeding each bootstrap worker independently is not equivalent to the serial computation: it moved Γ^* from 52.38 to 61.66 in one test and changed the cluster count from two to one. Advancing a single PCG64 stream to the offset each realisation would have reached serially reproduces the serial result exactly. Measured speedups are $1.67\times$ at $n_{\text{boot}} = 4$ and $3.47\times$ at $n_{\text{boot}} = 20$, with $\max_i |\Delta\Gamma_i| = 1.4 \times 10^{-14}$ against serial.

C Reserved

Reserved for the estimator implementation item of Section 2.3, pending discussion with the EagleEye authors.

D Figure manifest

All figures are symlinked into `report/figures/` from the run directories under `field_diagnostics/`, so the report is self-contained while the underlying products stay with their runs.

File	Section	Source run
fig1_reticulumII_sky.png	7.1	Ret II, $h=1.3333$, $K_M=100$, $p_{\text{ext}}=5\text{e-}4$
fig1b_reticulumII_pm.png	7.1	as above
fig2_bootesII_sky.png	7.2 , 5.3	Boötes II, $h=1.0$, $K_M=50$, $p_{\text{ext}}=1\text{e-}3$, eq. on
fig2b_bootesII_cmd.png	7.2	as above
fig3_scan_overlay.png	8.3	Ret II scan, gid 0, $p_{\text{ext}}=5\text{e-}3$, K_M auto
fig4_reticulumII_gamma_null.png	6.2	Ret II, as fig. 1

Figure 4 (the empirical Γ null against the analytic Fisher curves) is available and referenced by Section [6.2](#) but is not yet discussed in the text; adding that discussion is a small outstanding item.

References

- Battaglia G. et al., 2022, A&A, 657, A54
Darragh-Ford E., Nadler E. O., McLaughlin S., Wechsler R. H., 2021, ApJ, 915, 48
Drlica-Wagner A. et al., 2020, ApJ, 893, 47
d’Errico M., Facco E., Laio A., Rodriguez A., 2021, Information Sciences, 560, 476
Ester M., Kriegel H.-P., Sander J., Xu X., 1996, Proc. KDD-96, 226
Fisher R. A., 1932, Statistical Methods for Research Workers, 4th edn. Oliver & Boyd, Edinburgh
Gaia Collaboration, 2023, A&A, 674, A1
Glielmo A. et al., 2022, Patterns, 3, 100589
Guth A. H., Namjoo M. H., 2026, arXiv:2602.10178
Overdeck K. et al., 2026, arXiv:2606.09975
Rusterucci S., Hammer F., Yang Y., 2026, A&A, accepted (arXiv:2608.21517)
Springer S., Scaffidi A., Autenrieth M., Contardo G., Laio A., Trotta R., Haario H., 2025, Scientific Reports (arXiv:2503.23927)
Tan C. Y. et al., 2026, ApJ, 1000, 87
Vasiliev E., Baumgardt H., 2021, MNRAS, 505, 5978